Determine optimal worst-case distortion for ordinal min-cost metric perfect matching mechanisms

ChatGPT (gpt-5.6-sol)*

Partial progress generated on 2026-08-24

Abstract

We present partial progress on an open problem from *Distortion in metric matching with ordinal preferences*.

1 The open problem

This paper addresses an open problem posed by Nima Anari, Moses Charikar, Prasanna Ramakrishnan in *Distortion in metric matching with ordinal preferences* [1] (Economics and Computation (EC), 2023), Section 8 (Discussion), page 17, under problem 1 (also introduced as Question 1 in Section 1, page 2).. The website catalogue entry is problem 1 (`W4377866503_p0`) of the Open Problems in Operations Research collection.

> *Determine, as a function of n, the smallest achievable worst-case distortion among:*
>
> *1) deterministic ordinal matching mechanisms $\mathcal{M}$: (rankings) $\mapsto M$ that output a perfect matching, i.e.*
>
> $$\inf_{\mathcal{M}\ \textit{deterministic}} \ \sup_{\textit{profiles}} \ \sup_{d\ \textit{consistent}} \frac{\mathrm{cost}_d(\mathcal{M}(\textit{profile}))}{\mathrm{cost}_d(M_d^*)};$$
>
> *2) randomized ordinal matching mechanisms, i.e.*
>
> $$\inf_{\mathcal{M}\ \textit{randomized}} \ \sup_{\textit{profiles}} \ \sup_{d\ \textit{consistent}} \frac{\mathbb{E}[\mathrm{cost}_d(M)]}{\mathrm{cost}_d(M_d^*)}.$$
>
> *Equivalently, determine tight asymptotic bounds (up to constant factors, or otherwise as stated) on the optimal worst-case distortion for deterministic mechanisms and for randomized mechanisms in the minimum-cost metric perfect matching problem given only agents' ordinal rankings over items.*

The remainder of this document is the partial progress produced by the model, reproduced verbatim from the pipeline output.

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*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve → A.8 verify → A.10 repair write-up; run `run_20260824_024352_e5b85970, ec-2it-ultra`); two-iteration run, outcome from iteration 2. Independent verifier assessment: REJECTED with progress score 2/3 (major partial progress). Maintained by the AI in Mathematics & Operations Research project.

Overview and status

The status of this work is **partial progress, not a complete asymptotic solution**.

The verifier's repairable objection is fully resolved below: the serial-dictatorship recurrence is now derived, and it is in fact sharp for every fixed priority order. The fundamental objection remains: the arguments still do not determine D_n or R_n asymptotically.

Let D_n and R_n denote the optimal deterministic and randomized distortions. The strongest general bounds proved here are

$$\boxed{1 + 2\lfloor \log_2 n \rfloor \le D_n \le \min\{2^n - 1,\, nR_n\} \le \min\{2^n - 1,\, n^2\},}$$

and, for $k = \lfloor \log_2 n \rfloor$,

$$\boxed{k + \frac{n}{2^k} \le R_n \le n.}$$

The exact small cases remain

$$\boxed{(D_1, R_1) = (1, 1), \qquad (D_2, R_2) = (3, 2), \qquad (D_3, R_3) = \left(3, \frac{5}{2}\right).}$$

The unresolved general gaps are therefore

$$\Omega(\log n) \le R_n \le O(n), \qquad \Omega(\log n) \le D_n \le O(n^2).$$

2 Common-profile tree lower bounds

Let T be a full rooted binary tree with n leaves. Identify its leaves with the agents and its $n-1$ internal vertices with $n-1$ items, and add one further item r.

Fix a leaf s. For each ancestor U of s, let V be the child of U not containing s, and define

$$C_{s,U} = L(V) \cup I(V) \cup \{U\}.$$

Together with $\{s\}$ and $\{r\}$, these sets form a partition of all vertices. Moreover,

$$|L(V)| = |I(V)| + 1 = |I(V) \cup \{U\}|,$$

so every $C_{s,U}$ contains equally many agents and items.

Define a pseudometric d_s on the quotient classes by

$$d_s(s, C_{s,U}) = d_s(s, r) = 1,$$

and by assigning distance 2 to every other pair of distinct quotient classes. This is a metric on the quotient because its only nonzero distances are 1 and 2.

For a fixed agent $t \neq s$, let $U = \mathrm{LCA}(s,t)$, and let W be the child of U containing t. The items at distance zero from t are

$$Z_{s,t} = I(W) \cup \{U\}.$$

As s varies, these sets are nested; including the empty set arising from $s = t$, they form a chain of prefixes. Hence one strict profile is weakly compatible with every d_s.

Within each balanced class $C_{s,U}$, match agents to items at zero cost, and match s to r. This costs 1. Conversely, every assignment to r has cost at least 1, so

$$\mathrm{OPT}(d_s) = 1.$$

2.1 The deterministic adversary

Fix the matching M selected on the common profile. In a subtree U,

$$|L(U)| = |I(U)| + 1,$$

so some $t \in L(U)$ satisfies $M(t) \notin I(U)$. Mark such a t, descend into the child of U not containing t, and continue until a leaf s is reached.

The marked agents belong to disjoint discarded subtrees. Under d_s, each marked agent has its zero-distance items inside $I(U)$, whereas its assigned item lies outside $I(U)$; hence it pays 2. The final leaf s pays exactly 1. Therefore

$$C_{d_s}(M) \geq 1 + 2\,\mathrm{depth}(s).$$

Choosing a near-complete tree whose leaf depths are k and $k+1$, where $k = \lfloor \log_2 n \rfloor$, yields

$$\boxed{D_n \geq 1 + 2k.}$$

2.2 The randomized adversary

Let p be the doubly stochastic marginal matrix chosen on the common profile. At a reached subtree U, define

$$E_U = \sum_{\substack{t \in L(U) \\ j \notin I(U)}} p_{tj}.$$

Row and column sums give

$$E_U \geq |L(U)| - |I(U)| = 1.$$

Write X_L and X_R for the parts of this exit mass sourced in the two children, and choose the next child uniformly at random. If the path enters the left child, every right-sourced exit edge has final cost 2; if it enters the right child, every left-sourced exit edge has cost 2. Thus the conditional expected newly certified cost is

$$\frac{1}{2}(2X_L) + \frac{1}{2}(2X_R) = E_U \geq 1.$$

The discarded source sets are disjoint across levels, and the final leaf contributes exactly 1. Hence

$$\mathbb{E}_s\, C_{d_s}(p) \geq 1 + \mathbb{E}[\mathrm{depth}(s)].$$

Starting from a complete tree of depth k and splitting $n - 2^k$ of its leaves gives

$$\mathbb{E}[\mathrm{depth}(s)] = k + \frac{n - 2^k}{2^k}.$$

Therefore some s satisfies

$$C_{d_s}(p) \geq k + \frac{n}{2^k},$$

and consequently

$$\boxed{R_n \geq k + \frac{n}{2^k}.}$$

2.3 Strict genuine metrics

Choose the common strict profile and let $r_i(j)$ be the rank of item j for agent i. Define

$$\rho(a_i, b_j) = 1 + \frac{r_i(j)}{n},$$

and set $\rho(x, y) = 2$ for distinct vertices on the same side. Since every cross-distance lies in $[1, 2)$, the function ρ is a genuine metric inducing the strict profile.

Then

$$d_{s,\varepsilon} = d_s + \varepsilon\rho$$

is a genuine metric inducing the same profile. As $\varepsilon \downarrow 0$, finiteness of the set of matchings gives

$$\mathrm{OPT}(d_{s,\varepsilon}) \longrightarrow 1, \qquad C_{d_{s,\varepsilon}}(p) \longrightarrow C_{d_s}(p),$$

so the lower bounds hold for genuine strict metrics.

3 The binary-tree construction is already saturated

The preceding construction cannot be amplified merely by using an unbalanced tree or different random exporter choices.

Construct a tree-respecting matching recursively: each subtree exports one unmatched leaf-agent; at an internal node, match the node's item to the exporter of one child and let the other child's exporter survive; at the root, match the final exporter to r.

For a leaf s, let $L(s)$ be the number of ancestors at which the child containing s is the consumed rather than the surviving child. A direct induction on the tree gives

$$C_{d_s}(M) = 1 + 2L(s).$$

Let $h(T)$ be the minimum, over deterministic orientations, of $\max_s L(s)$. If the children have values h_L and h_R, then

$$h(T) = \min\{\max(h_L + 1, h_R),\ \max(h_L, h_R + 1)\},$$

or equivalently

$$h(T) = \begin{cases} \max(h_L, h_R), & h_L \neq h_R, \\ h_L + 1, & h_L = h_R. \end{cases}$$

This is the Horton–Strahler recurrence. A tree with $h(T) \geq q$ has at least 2^q leaves, so

$$h(T) \leq \lfloor \log_2 n \rfloor.$$

Thus every n-leaf binary tree admits a single matching whose cost is at most

$$1 + 2\lfloor \log_2 n \rfloor$$

simultaneously on all of its special metrics; near-complete trees attain equality.

For randomized orientations, let $v(T)$ be the minimum possible value of $\max_s \mathbb{E}\, L(s)$. If the left child survives with probability q, then

$$v(T) = \min_{q\in[0,1]} \max\{v_L + 1 - q,\ v_R + q\},$$

hence

$$v(T) = \max\left\{v_L,\, v_R,\, \frac{1 + v_L + v_R}{2}\right\}.$$

Induction on the number of leaves yields

$$\max_{|L(T)|=n} v(T) = \frac{1}{2}\left(k - 1 + \frac{n}{2^k}\right), \qquad k = \lfloor \log_2 n \rfloor.$$

Consequently, the best randomized exporter scheme has worst special-metric cost

$$1 + 2v(T) \le k + \frac{n}{2^k},$$

again with equality for a near-complete tree. Thus the deterministic and randomized lower bounds are the exact minimax values of this entire one-exporter binary-tree template.

4 A broader barrier for one-scale lower bounds

The logarithmic barrier extends well beyond laminar trees.

Let d be any compatible pseudometric whose nonzero agent–item distances lie in $[L, U]$, where $0 < L \le U$. Let C range over the zero-distance equivalence classes, and write

$$A_C = |A \cap C|, \qquad B_C = |B \cap C|, \qquad E = \sum_C (A_C - B_C)_+.$$

Every perfect matching has at least E cross-class edges, so

$$\mathrm{OPT}(d) \ge LE.$$

Run random serial dictatorship (RSD). With r agents remaining, let $A_C(t)$ and $B_C(t)$ denote the residual counts, and put

$$E_t = \sum_C (A_C(t) - B_C(t))_+.$$

An agent crosses classes only if its own class has no item remaining, since all of its zero-distance items form a ranking prefix. The number of currently exposed agents is therefore at most E_t.

A within-class assignment leaves every imbalance unchanged, while a cross-class assignment decreases the positive imbalance of its source class by one and raises the positive imbalance of its destination class by at most one. Hence

$$E_t \le E$$

throughout. Conditional on the history,

$$\Pr(\text{next edge crosses} \mid r) \le \min\left\{1, \frac{E}{r}\right\}.$$

Therefore

$$\mathbb{E}[\#\text{ cross edges}] \le \sum_{r=1}^{n} \min\left\{1, \frac{E}{r}\right\} = E\bigl(1 + H_n - H_E\bigr),$$

where $H_m = \sum_{j=1}^{m} 1/j$. It follows that

$$\boxed{\frac{\mathbb{E}\, C_d(\mathrm{RSD})}{\mathrm{OPT}(d)} \le \frac{U}{L}\bigl(1 + H_n - H_E\bigr) \le \frac{U}{L} H_n.}$$

In particular, for all 0/1/2-valued compatible pseudometrics,

$$\mathrm{distortion}(\mathrm{RSD}) \le 2H_n.$$

Thus no common-profile mixture consisting only of 0/1/2 metrics—laminar or crossing, with one exporter or many—can prove a randomized $\omega(\log n)$ lower bound.

There is also a multiscale extension. Suppose

$$d(x, y) = \sum_{\ell} w_\ell\, \mathbf{1}\{P_\ell(x) \ne P_\ell(y)\}, \qquad w_\ell \ge 0,$$

where each P_ℓ is a partition and, for every agent, the items in its P_ℓ-block form a ranking prefix. Applying the imbalance argument to each layer gives

$$\mathbb{E}\, C_d(\mathrm{RSD}) \le H_n \sum_{\ell} w_\ell E_\ell \le H_n\, \mathrm{OPT}(d).$$

Every finite ultrametric has exactly such a nested-partition representation. Hence

$$\boxed{\mathrm{distortion}_{\mathrm{ultrametric}}(\mathrm{RSD}) \le H_n.}$$

This does not extend automatically to general metrics: positive-radius balls can overlap rather than form partitions. The smallest obstruction occurs on the line with items $x = 0$, $y = 2$ and agents at 1 and 0. At radius 1, the first agent can use either item, while the second can use only x. Although the threshold graph has a perfect matching, if the flexible agent moves first and selects x, the second agent is forced outside its threshold. The conserved component-imbalance argument depends essentially on equivalence classes, not merely on prefix neighborhoods.

Therefore any superlogarithmic lower-bound construction must exploit genuinely nonhierarchical, multiscale geometry.

5 A general randomized upper bound

We use random serial dictatorship. Consider a residual instance of size m and any reference matching O, with

$$C = \sum_i x_i, \qquad x_i = d(a_i, O_i).$$

The first random agent i chooses its favorite remaining item g_i, so

$$d(a_i, g_i) \le x_i.$$

If $g_i = O_k$ with $k \ne i$, delete i and g_i, and assign O_i to agent k in the residual reference matching. The triangle inequality gives

$$d(a_k, O_i) \le d(a_k, O_k) + d(a_i, O_k) + d(a_i, O_i) \le x_k + 2x_i.$$

More sharply, retaining $d(a_i, g_i)$ in the calculation, the residual reference cost is at most

$$C + d(a_i, g_i) \le C + x_i.$$

The same bound is immediate when $k = i$. Averaging over the uniformly random first agent,

$$\mathbb{E}[d(a_i, g_i)] \le \frac{C}{m}, \qquad \mathbb{E}[C_{\text{res}}] \le C + \frac{C}{m}.$$

If RSD on $m-1$ agents costs at most $m-1$ times the cost of every reference matching, then

$$\mathbb{E}\, C_{\text{RSD}} \le \frac{C}{m} + (m-1)\left(C + \frac{C}{m}\right) = mC.$$

Thus

$$\boxed{R_n \le n.}$$

All inequalities are cross-multiplied, so the proof remains valid when $C = 0$.

6 Derandomization

Let p be the marginal matrix of any randomized ordinal rule. Write agent i's ranking as

$$b_{i1} \succ \cdots \succ b_{in}, \qquad q_i(k) = \sum_{r=1}^{k} p_{i,b_{ir}},$$

and let

$$k_i = \min\left\{k : q_i(k) > 1 - \frac{1}{n}\right\}.$$

Join each agent i to its first k_i items. The resulting graph has a perfect matching: otherwise, Hall's theorem gives a set S with $T = N(S)$ and $|T| \le |S| - 1$; but then

$$p(S,T) > \left(1 - \frac{1}{n}\right)|S| \ge |S| - 1,$$

whereas the column capacities give

$$p(S,T) \le |T| \le |S| - 1,$$

a contradiction.

Choose a perfect matching M in this prefix graph. For a compatible metric, put

$$z_i = \sum_j p_{ij}\, d(a_i, b_j).$$

Minimality of k_i places probability at least $1/n$ on rank k_i or worse, so

$$z_i \ge \frac{1}{n}\, d(a_i, b_{i,k_i}),$$

while $M(i)$ lies among the first k_i positions, whence

$$d(a_i, M(i)) \le d(a_i, b_{i,k_i}) \le n z_i.$$

Summing over the agents,

$$C_d(M) \le n\, \mathbb{E}\, C_d(p).$$

Applying this to rules of distortion $R_n + \varepsilon$ and letting $\varepsilon \downarrow 0$ yields

$$\boxed{D_n \le nR_n \le n^2.}$$

7 The repaired fixed-order serial-dictatorship recurrence

This section resolves the verifier's repairable objection.

Let α_m be the worst-case distortion of serial dictatorship with a fixed priority order on m agents. Let $e = (i, g)$ be its first edge, and let O be an optimal matching, of cost C. Since g is agent i's favorite item,

$$d(e) \le d(i, O_i) \le C.$$

If $g = O_k$, splice O after deleting i and g by assigning O_i to k. As above, the residual reference cost is at most

$$C + d(e) \le 2C.$$

Therefore

$$C_{\mathrm{SD}} \le d(e) + \alpha_{m-1}(2C) \le (1 + 2\alpha_{m-1})C.$$

Hence

$$\boxed{\alpha_1 = 1, \qquad \alpha_m \le 1 + 2\alpha_{m-1},}$$

and therefore

$$\boxed{\alpha_m \le 2^m - 1.}$$

This recurrence is sharp for every fixed priority order. Place the agents on the line at

$$a_i = 2^{i-1},$$

put

$$b_i = 2^i \quad (i < n), \qquad b_n = -\varepsilon,$$

and use the priority order $a_1, \dots, a_n$. Serial dictatorship assigns a_i to b_i for $i < n$ and a_n to b_n, at cost

$$2^n - 1 + \varepsilon.$$

The matching

$$a_1 \mapsto b_n, \qquad a_i \mapsto b_{i-1} \quad (i \ge 2)$$

costs $1 + \varepsilon$. Letting $\varepsilon \downarrow 0$ gives distortion $2^n - 1$.

Thus the missing derivation is fully repaired, and the bound cannot be improved for a fixed, profile-independent priority order.

8 Exact values for $n \le 3$

For $n = 2$, if both agents have the same top item, write

$$x = d(a_1, b_1) \le y = d(a_1, b_2), \qquad u = d(a_2, b_1) \le v = d(a_2, b_2).$$

The two matching costs are $C_1 = x + v$ and $C_2 = y + u$. The triangle inequality gives

$$v \le u + x + y, \qquad y \le x + u + v,$$

and hence

$$C_1 \le 3C_2, \qquad C_2 \le 3C_1.$$

The uniform lottery costs at most twice the cheaper matching. The two compatible pseudometrics

$$\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$$

give ratios $1 + 2q$ and $3 - 2q$ when the first matching is selected with probability q. Thus

$$D_2 = 3, \qquad R_2 = 2.$$

For D_3, call a matching *strongly popular* if no comparator matching improves two agents. Every non-unanimous profile has such a matching:

- If at least two top items are distinct, assign two distinct top items and complete the matching.
- If all top choices coincide but the second choices do not, every agent can be assigned its first or second choice; two agents cannot both improve, because both would require the unique common top item.

Against any comparator, each alternating component then has at most one agent who prefers the comparator. For a two-cycle with bad agent b and good agent g,

$$m_b \le o_b + m_g + o_g \le o_b + 2o_g, \qquad m_g \le o_g,$$

so its cost is at most three times the comparator cost. For a three-cycle,

$$m_b \le o_b + m_{g_1} + o_{g_1} + m_{g_2} + o_{g_2} \le o_b + 2o_{g_1} + 2o_{g_2},$$

and adding the good-agent inequalities again gives the factor 3.

For the unanimous profile $x \succ y \succ z$, relabel so that $M = (x, y, z)$. Only the comparator $O = (z, x, y)$ can improve two agents. Writing $m_i = d(a_i, M_i)$ and $o_i = d(a_i, O_i)$,

$$m_0 \le o_0, \qquad m_1 \le o_1 + 2o_2, \qquad m_2 \le o_2 + 2o_0.$$

Therefore $C(M) \le 3C(O)$. Together with the tree lower bound, this gives

$$D_3 = 3.$$

For R_3, the top-choice patterns are $(1, 1, 1)$, $(3, 0, 0)$, and $(2, 1, 0)$.

In the first pattern, matching all top choices is optimal. In the common-top case, the uniform assignment has cost at most $7C/3$: if h is the common top item and $O(k) = j$, then

$$\sum_i d(i, h) \le C$$

and

$$\sum_i d(i, j) \le C + 4\, d(k, j).$$

Summing the three columns gives at most $7C$.

For the $(2, 1, 0)$ pattern, let u, v top-rank x, let w top-rank y, and let z be unclaimed. Define

$$P(i; a, b) = d(i, b) - d(i, a) \ge 0$$

whenever i ranks a above b, and

$$Q(i,j;k,\ell)=d(i,k)+d(\ell,k)+d(\ell,j)-d(i,j)\geq 0.$$

If u and v both rank $x\succ y\succ z$, use

$$q_u=q_v=\tfrac{1}{2}x+\tfrac{1}{8}y+\tfrac{3}{8}z,\qquad q_w=\tfrac{3}{4}y+\tfrac{1}{4}z.$$

For the six comparator matchings, direct expansion writes $\alpha C(O)-C(q)$ as a nonnegative combination of the quantities P and Q and of distances, with

$$\alpha=\frac{5}{2},\frac{5}{2},\frac{5}{2},\frac{9}{4},\frac{5}{2},\frac{9}{4}$$

for

$$(x,y,z),\ (x,z,y),\ (y,x,z),\ (y,z,x),\ (z,x,y),\ (z,y,x),$$

respectively. Hence $C(q)\leq\frac{5}{2}C(O)$.

If one contender, relabelled u, ranks $x\succ z\succ y$, use

$$q_u=q_v=\tfrac{1}{2}x+\tfrac{1}{2}z,\qquad q_w=y.$$

The same six expansions give the factors

$$2,\ 2,\ 2,\ 1,\ 2,\ 2,$$

respectively. Every term used is one of the ranking slacks P, the triangle slacks Q, or a nonnegative distance, so the inequalities are valid for every compatible metric. Thus this case has distortion at most 2.

Together with the three-leaf tree lower bound,

$$R_3=\frac{5}{2}.$$

9 An exact minimax reformulation of the remaining problem

For a fixed profile σ, let K_σ be the cone of nonnegative cross-cost matrices satisfying the ordinal inequalities together with

$$c_{ij}\leq c_{i\ell}+c_{k\ell}+c_{kj}\qquad\forall\, i,k,j,\ell.$$

These are exactly the agent–item restrictions of pseudometrics. Necessity follows from the triangle inequality along

$$a_i-b_\ell-a_k-b_j.$$

For sufficiency, place edge length c_{ij} on the complete bipartite graph and take the shortest-path distance: every longer odd path can be shortened three edges at a time using the displayed inequality, so the direct cross edge remains shortest.

Let

$$S_\sigma=\{c\in K_\sigma:\mathrm{OPT}(c)=1\}.$$

Then the profilewise optimal randomized distortion is exactly

$$\boxed{R(\sigma)=\sup_{\bar{c}\in\operatorname{conv}S_\sigma}\mathrm{OPT}(\bar{c}).}$$

Indeed,

$$R(\sigma) = \min_{p\in\mathcal{B}_n} \sup_{c\in S_\sigma} p\cdot c.$$

Convexifying the adversary and applying the minimax theorem gives

$$R(\sigma) = \sup_{\bar{c}\in\operatorname{conv} S_\sigma} \min_{p\in\mathcal{B}_n} p\cdot \bar{c}.$$

By Birkhoff–von Neumann, the inner minimum is the cost of an optimal perfect matching for $\bar{c}$. Truncation and a limiting argument handle the unbounded cone, and Carathéodory shows that an approximately worst mixture needs at most n^2+1 metrics.

Thus the randomized problem is precisely the following Jensen-gap question:

If $c^1,\dots,c^m$ *are co-ordinal metric cost matrices with* $\mathrm{OPT}(c^s)=1$*, how large can*

$$\mathrm{OPT}\left(\sum_s \lambda_s c^s\right)$$

be?

The tree construction gives a logarithmic Jensen gap. A universal $O(\log n)$ bound would close the randomized asymptotics; a superlogarithmic mixture would improve the lower bound. Neither is proved here.

In this formulation one must allow genuinely new hybrid matchings. For example, the co-ordinal matrices

$$c=\begin{pmatrix}0&0&2&2\\0&0&2&2\\2&2&0&2\\1&1&1&1\end{pmatrix},\qquad d=\begin{pmatrix}1&1&1&1\\0&2&2&2\\2&0&0&2\\2&0&0&2\end{pmatrix}$$

have respective optimal matchings

$$O=(0,1,2,3),\qquad Q=(3,0,1,2)$$

of cost 1, but

$$d(O)=c(Q)=5.$$

Nevertheless,

$$\mathrm{OPT}\left(\frac{c+d}{2}\right)=2,$$

attained by $M=(1,0,2,3)$. A dual certificate is

$$u=\left(\tfrac{1}{2},1,0,\tfrac{1}{2}\right),\qquad v=(-1,0,0,1),$$

for which $u_i+v_j\le(c_{ij}+d_{ij})/2$ and $\sum_i u_i+\sum_j v_j=2$. Hence merely selecting one scenario's optimum cannot prove the desired upper bound.

10 Why the obvious recursive repair still fails

For any partial matching P and perfect matching O, the remaining vertices admit a perfect matching N with

$$C(N) \le C(O) + C(P).$$

To see this, decompose $P \cup O$ into alternating cycles and paths. Every path joins an unmatched agent to an unmatched item; shortcut each path using the triangle inequality.

After m RSD choices from an n-agent instance,

$$\frac{\mathbb{E}\, C(P_m)}{\mathrm{OPT}} \le \frac{m}{n-m+1}.$$

If one stops with k agents remaining and then invokes an r_k-distortion subroutine, this gives only

$$r_n \le \frac{n-k+(n+1)r_k}{k+1}.$$

For $k \approx n/2$, the recurrence has the form

$$r_n \lesssim 1 + 2r_{n/2},$$

which remains linear, rather than the needed $1 + r_{n/2}$.

Probabilistic Serial does not repair the gap either. On the line, put

$$a_i = 2^{i-1}, \qquad b_1 = -1, \qquad b_i = 2^{i-1}\ (i \ge 2).$$

The optimum costs 2. Under Probabilistic Serial, item b_i is exhausted at time $1 - 2^{-(i-1)}$; the displaced consumption of $b_2, \dots, b_n$ costs $(n-1)/2$, while the consumption of b_1 costs $(n+3)/2$. Its total cost is therefore $n+1$, giving distortion

$$\frac{n+1}{2}.$$

Therefore neither truncated RSD, nor Probabilistic Serial, nor unbalanced exporter trees, nor crossing one-scale zero partitions, nor ultrametric multiscale refinements closes the general gap.

The fixed-order recurrence objection is fully repaired. The fundamental asymptotic objection remains unresolved: the strongest rigorously proved general results are still the displayed $\Omega(\log n)$-to-$O(n)$ and $\Omega(\log n)$-to-$O(n^2)$ bounds, together with the exact values through $n = 3$ and the logarithmic upper bounds for the restricted quotient and ultrametric classes.

References

[1] Nima Anari, Moses Charikar, Prasanna Ramakrishnan. *Distortion in metric matching with ordinal preferences.* Economics and Computation (EC), 2023. https://doi.org/10.1145/3580507.3597740